

Lecture 8: Total Probability

24 June 2026 12:30

Topics of this lecture:

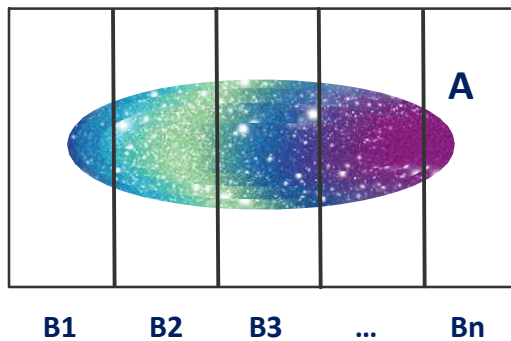
1. Total Probability

1. Total Probability:

Total probability is the way of relating marginal probabilities to conditional probabilities

Total probability of an event is expressed with multiple distinct events.

We have to find the probability of event A; but the sample space B is divided into multiple parts exactly like shown in the figure.



$$B = \{B_1 \cup B_2 \cup B_3 \dots \cup B_n\}$$

Here sample space B is set of partwise disjoint events whose union is entire sample space

Here probability of any event $P(A) = \sum P(A \cap B_n)$

$$P(A) = \sum P(A | B_n) \cdot P(B_n)$$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + \dots + P(A \cap B_n)$$

$$P(A) = P(A | B_1) \cdot P(B_1) + P(A | B_2) \cdot P(B_2) + P(A | B_3) \cdot P(B_3) + \dots + P(A | B_n) \cdot P(B_n)$$

Experiment 1: $P(\text{Rain}) = 60\%$

$P(\text{Not Rain}) = 40\%$

$P(\text{School} \mid \text{Rain}) = 10\%$

$P(\text{School} \mid \text{Not Rain}) = 80\%$

$P(\text{School}) = ?$

By applying Total Probability Formula

Probability of going to school is:

$$P(\text{School} \mid \text{Rain}) \cdot P(\text{Rain}) + P(\text{School} \mid \text{Not Rain}) \cdot P(\text{Not Rain})$$

$$= 0.10 \times 0.60 + 0.80 \times 0.40$$

$$= 0.06 + 0.32$$

$$= 0.38$$

$$P(\text{School}) = 38\%$$